



KALMAN

Machinable Aluminum Nitride

APPLICATIONS

- Heat sinks
- Electrical insulation components
- Vacuum components
- High-temperature precision-machined parts

MAIN PROPERTIES

Property	Value	Unit
Density	3.2 – 3.3	g/cm ³
Thermal Conductivity	140 – 170	W/m·K
CTE	2.2 – 2.6	x10 ⁻⁶ K ⁻¹
Thermal Conductivity (100°C)	>80	W/m·K

* All properties measured at 20°C unless otherwise stated.

** Measured in a wall thickness of 2 mm

OPERATING CONDITIONS

Max working temp (air)	1000	°C
Max working temp (inert)	1900	°C
Recomended temp	1350	°C
Continuous temp	1350	°C

ELECTRICAL BEHAVIOR

Resistivity (25°C)	>1E12	Ω·cm
Resistivity (1000°C)	1E6 - 1E8	Ω·cm
Resistivity (1300°C)	1E4 - 1E6	Ω·cm
Dielectric strength (50° Hz)	>18	kV/mm

MATERIAL INFORMATION

Material type	AlN-based
Machinability	High
Compability (Mo)	Yes
Structure	Machinable ceramic
Purity	High

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

The information provided is for guidance only and does not constitute a guarantee.

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